INCH-POUND

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May 7, 2025
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SEALING COMPOUND (FAIRING COMPOUND), POLYSULFIDE, FOR WATERBORNE MAIN PROPULSION SHAFTING COVERINGS ON U.S. NAVAL SURFACE SHIPS AND SUBMARINES

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) describes the polysulfide fairing/sealing compound that covers and protects all shaft-sleeve interface areas of waterborne main propulsion shafting by preventing galvanic corrosion to the steel substrate in the marine environment and so preventing structural failure of the shafting. This CID pertains specifically to the fourth component of the five-component shaft covering system specified in MIL-STD-2199.

2. CLASSIFICATION. The layers shall conform to the following type:

a. Type I: Deleted.
b. Type II: Deleted.
c. Type III: Fourth component (fairing/sealing compound). High-viscosity, thixotropic elastomeric coating.

3. SALIENT CHARACTERISTICS.

3.1 Fourth component (fairing/sealing compound). The fourth component of the shaft covering system shall be an elastomeric fairing/sealing compound, specifically, a polysulfide with a mixed viscosity of 700,000 centipoise or greater (Thiokol 2282, or equal). The fairing/sealing compound shall consist of a 100 percent solid, high-viscosity, thixotropic elastomeric coating for fairing the shaft to the sleeve and as a filler material for surface irregularities in the shaft. The fairing/sealing compound shall bond to the ultra-high solids (UHS) coating and to the glass reinforced plastic (GRP) shell specified in MIL-STD-2199.

3.1.1 Adhesion of fairing/sealing compound to UHS Coating. The fairing/sealing compound shall be adhered to MIL-PRF-23236, type VII, UHS epoxy novolac amine coatings with the following: 10 to 12 mils of Sherwin-Williams Nova-Plate® UHS Primer (B62L220-part A and B62V221-part B, or equal) followed by 10 to 12 mils of Sherwin-Williams Nova-Plate® UHS (B62W220-part A and B62GV221-part B, or equal). Adhesion of the fairing/sealing compound to the UHS coating shall be tested both dry and after 1 year of immersion in seawater in accordance with ASTM D1141. Adhesion shall be determined in accordance with ASTM D4541 with the P.A.T.T.I.®, or equal, portable tester applied to the GRP or in accordance with MIL-R-23461. The second and third components of MIL-STD-2199 shall be applied to 12- by 12- by ¼-inch flat steel test panels in accordance with MIL-R-23461 that have been white metal blasted to a 3-mil average profile in accordance with SSPC-SP 5/NACE No. 1. Primed and coated test panels shall be covered with a 20±5-mil (0.020±0.005-inch) layer of fairing/sealing compound after the UHS coating cures. After the fairing/sealing compound cures, the test panel shall be covered with four plies of GRP applied in accordance with MIL-STD-2199. Both dry and wet fairing/sealing

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compound-to-UHS coating adhesive strength values shall exceed 400 pounds per square inch (psi), and any failure shall occur cohesively in the fairing/sealing compound. Adhesive failure at the fairing/sealing compound-to-UHS coating bond shall constitute rejection of the fairing/sealing compound.

3.1.2 Adhesion of GRP to fairing/sealing compound. Adhesion of GRP to the fairing/sealing compound shall be tested both dry and after 1 year of immersion in seawater in accordance with ASTM D1141. The primed and coated test panels specified in 3.1.1 shall be covered with a 20±5-mil (0.020±0.005-inch) layer of fairing/sealing compound after the coating cures. After the fairing/sealing compound cures, the test panel shall be covered with four plies of GRP applied in accordance with MIL-STD-2199. GRP adhesion to the fairing/sealing compound shall be determined in accordance with ASTM D4541 using a portable adhesion tester such as the P.A.T.T.I.®, or equal. The P.A.T.T.I.® tester, or equal, shall be applied to the GRP. Both dry and wet GRP-to-fairing/sealing compound adhesive strength values shall exceed 400 psi, and any failure shall occur cohesively in the fairing/sealing compound. Adhesive failure at the GRP-to-fairing/sealing compound bond shall constitute rejection of the fairing/sealing compound.

3.1.3 Cure characteristics. The fairing/sealing compound shall achieve full cure at ambient room temperature. The pot life shall be a minimum of 30 minutes.

3.1.4 Curing constituents. The polysulfide components of the fairing/sealing compound shall be free of solvents and lead oxide curing agents.

3.1.5 Application characteristics. The fairing/sealing compound shall be capable of being applied to rotating shaft sections using a trowel or squeegee. The viscosity or thixotropy shall allow the material to fair the shaft to the sleeve end.

3.1.6 Solvent content. The fairing/sealing compound system shall be free of solvents and volatile organic compounds.

3.1.7 Tensile properties. Tensile elongation of the cured fairing/sealing compound shall exceed 400 percent at failure when tested in accordance with ASTM D638.

3.2 Toxicity. The fairing/sealing compound shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to this requirement as specified (see 7.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the fairing/sealing compound using data provided by the manufacturer/distributor to the NMCFHPC (see 7.5).

4. REGULATORY REQUIREMENTS.

4.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4.2 Environmental Protection Agency (EPA) compliancy. The amount of volatile organic compounds released shall be in accordance with the limits established by the EPA for shipyards in 40 CFR 63.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered shall have been previously sold either to the government or on the commercial market and shall currently remain a standard (currently produced) offering of the manufacturer. A currently produced item is required to help ensure up-to-date technology, especially in light of probable multiple-year contract durations and rapid advancements experienced in the technology.

6. PACKAGING. Preservation and packing shall be as specified in the contract or order (see 7.2). The product shall be marked as "Propulsion Shaft Fairing/Sealing Compound".

7. NOTES.

7.1 Source of documents.

7.1.1 ASTM standards. ASTM standards are available from www.astm.org.

7.1.2 Code of Federal Regulations (CFR). The CFR is available from www.ecfr.gov.

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7.1.3 Defense specifications and standards. Defense specifications and standards are available online at https://quicksearch.dla.mil.

7.1.4 SSPC/NACE standards. SSPC/NACE standards are available from www.ampp.org.

7.2 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Requirements for an HHA (see 3.2).
c. Product conformance provisions (see 5.1).
d. Packaging requirements (see 6).
e. When Safety Data Sheets (SDS) are required (see 7.4).
f. Quantity required.

7.3 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances in accordance with the conversion tables specified in FED-STD-376 and all other requirements of this CID are met, including form, fit, and function. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

7.4 SDS. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313, as specified (see 7.2). The pertinent government mailing addresses for submission of data are listed in FED-STD-313. 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation be readily available to personnel using the material.

7.5 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

7.6 Key words.

Elastomer
Flexible sealants
Polysulfide coatings
Polysulfide sealant

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – MR
Navy – SH

Preparing activity:
Navy – SH
(Project 9330-2025-002)

Review activities:
Navy – AS, CG

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